

Network Analysis of Crypto Currencies

Charles Bryan Dixon III
Collin Hunt

April 30, 2023

Abstract

We take a time series of 58 different Crypto currencies, get the covariance between them, and run some basic network analysis algorithms on them. The goal is to use the information to create a portfolio of Crypto assets that are diversified. We may not have the time to do as much as we desire, but the initial work will go a long way to those wanting to continue the research.

For data and code retrieval, please email dxnstatistics@gmail.com for permission to pull code from <https://github.com/DXNSTATISTICS>.

1 Introduction

Cryptocurrencies have differences, but they are all part of a network. Each cryptocurrency is associated with a blockchain that serves as its open ledger. For instance, a crypto currency like Tether runs on the Ethereum block chain. Another example, Litecoin, runs on the Bitcoin blockchain. So there is a technical network that exists in the cryptocurrency space on dependants and foundational blockchains. The other network is more of an indirect social network. The investors in the crypto space often invest in other coins. When they are bearish in general, they sell crypto assets, when they are bullish, the opposite is the case. So, it can be assumed that if crypto investors follow that understanding, a drop in price of one crypto can have an effect on others. This is why we took it upon ourselves to analyze the covariance matrix of cryptocurrencies using network analysis methods. The goal of this paper is not a specific one, but a general search for insights using methods of statistical network analysis. These insights can be used to raise new questions for further research.

2 Data and Methods

The data were collected using application programming interfaces (APIs) from TradingView.com. This was acquired through an unofficial Python API "TVDataFeed". The data acquired were time series data for a variety of cryptocurrencies. This analysis only compiled daily price data for 59. The data description can be found in Table 1.

The only variable of interest was the daily price changes. After the data were acquired, the variables 'open', 'high', 'low', and 'volume' are deleted from the data frame. We kept 'close' as our variable

Label	Description
'datetime'	The date (Year-Month-Day) of the data row
'open'	Opening price of asset that day.
'high'	Highest daily price of the asset.
'low'	Lowest daily price of the asset.
'close'	Closing price of asset that day.
'volume'	Amount traded that day.

Table 1: Data Description

of interest. The first three of these variables have near perfect correlation with 'close', our response variable. Volume was eliminated because it was not a variable of interest in this analysis.

That data was then combined into a dataframe "dfALL", which was then used to create a correlation and covariance matrix. We ran some analysis of the correlation, but determined that covariance was better for the analysis we wanted to do; though, we do not discount analysis of correlation being a viable topic of interest. In asset trading, like stocks and bonds, the covariance is used in measuring risk. A portfolio does not need to be heavily exposed to a single risk factor, so diversification between assets is done to lower the internal covarainces.

We deleted the covariance values of crypto currencies with themselves, which was just the assets variance, and we changed the "Index" value, which was a numeric place holder for the currency to the other, and changed it to string values of their ticker. Examples Include "BTC", "ETH", "MIM" etc. A complete list of crypto assets and their tickers can be found in the appendix. We also eliminated covariance between positive 5000 and negative 5000. We wanted to capture both positive and negative effects and eliminate the complexity when we include covariances close to zero. This 5000 value is an arbitrary number picked by viewing multiple graphs.

The crypto assets themselves are broke up into categories. These are "Currencies", "L1/L2 Platforms", "Decentralized Finance", "Web 3.0", and "Aggregators/Metaverse". These categories deserve more analysis, but for the sake of time and space, the analysis is not included in this paper. We also ran K-Means with K set to 5. This value of 5 is again aritrary. We tried a few other K values, but the graph did not adjust much beyonds its final product. We used KMeans to group some of these coins together.

The range of dates tested for the Cryptocurrencies in the analysis falls between 02/28/2022 - 02/23/2023. This is a very short time frame where prices were generally declining, then beginning to trend upwards recently. This was all the data we could get that included all the crypto currencies analyzed without difficulty. There are methods to get more data, or we can eliminate crypto assets that have only been around for a short period of time so that we can test the covariance of assets that have been around much longer, but for the sake of time we stuck with the data compiled.

3 Network Analysis Tests

We then calculated the degree of each individual node. We use this value to determine the importance of the node itself. It indicates how many connections the node has or neighbors in the network. In our calculation, Bitcoin is more than double the next largest degree node which is Ethereum. A surprise value is KP3R which has the third highest degree. This could be a coincidence based on the price changes over the last year or could be an interesting consideration that needs to be further analyzed. KP3R is not a big coin, which is why its degree is interesting.

Crypto Asset	Degree	Crypto Asset	Degree
BCH	4	BTC	35
LTC	2	MIM	2
PAXG	4	ZEC	3
ETH	13	KP3R	7
LUNA	3	QNT	1
SOL	2	XTZ	1
AAVE	3	ALCX	2
BAL	1	BNT	1
BOND	1	COMP	2
CRV	1	CTX	1
INJ	1	KNC	1
MCO2	1	API3	1
AXS	2	ENS	1
FIL	1	LINK	1
LPT	1	MASK	1
MC	1	RAD	1
RNDR	1	CUBE	1
MANA	1	SAND	1

Table 2: Degrees of Individual Nodes

The graph itself shows this information in a quality visualization. Bitcoin is obviously the most important node. It was the first crypto to market. The next is Ethereum, and then KP3R. KP3R may be a fluke of higher trading during this period because in general, the most highly traded cryptos are Ethereum and Bitcoin. If we run K-Means with $k=4$ and not 5, then KP3R will fall into the next group of coins (Orange Group). I do plan to buy some KP3R because of this result. The prices of all crypto currencies are depressed due to the market collapse AKA "Crypto winter", but if KP3R is a secretly popular coin, when the market turns around KP3R may increase faster than most other crypto assets.

The Orange group of coins have higher covariances to Ethereum and Bitcoin that include MIM, BCH, PAXG, ZEC, LUNA, LTC, ALCX, AXS, SOL, and AAVE. The final (Pink Group) consist of the majority of the crypto assets that have little covariance to Bitcoin. They may have correlation to others, but the covariance is too small to make it through our screen and this may be insignificant.

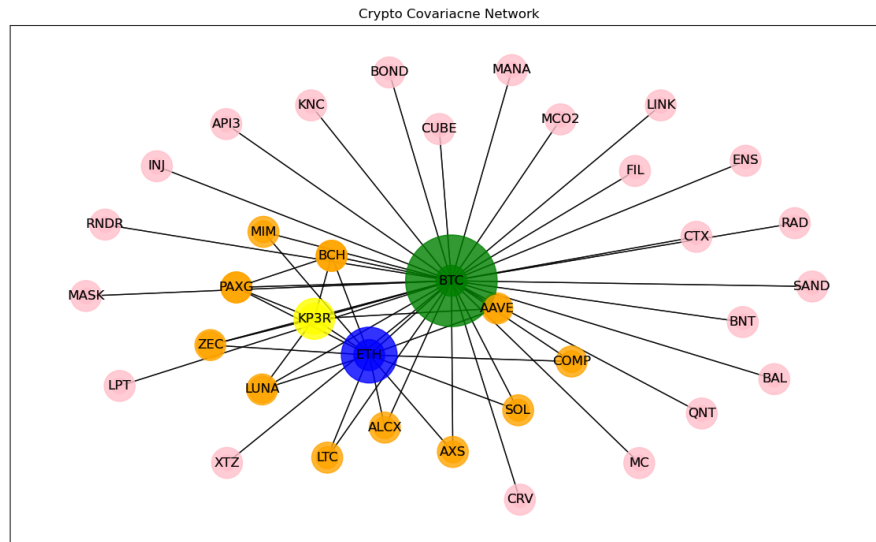


Figure 1: Crypto Covariance Network

4 Conclusion

Measuring covariance is a major step in reducing risk in a portfolio. In a crypto portfolio, the goal should be to diversify risk and increase reward. If you want to invest in some of the less mainstream crypto currencies, one must understand their covariance to the major crypro assets. Our analysis showed us what we already expected, Bitcoin and Ethereum are major nodes, but it gave us an unexpected result with Keep3rV1. Our analysis also helped us better group these crypto assets into various neighborhoods that will merit further research.

Future research will involve comparing these covariance groups with their overall return on assets. The goal is to increase the potential return while trying to keep a covariance value close to zero so huge negative market moves will not eliminate the entire portfolio.

5 Appendix

5.1 Crypto Asset Lists By Category

	Crypto Asset	Category
1	Bitcoin (BTC)	Currency
2	Bitcoin Cash(BCH)	Currency
3	Dai (DAI)	Currency
4	Dogecoin (DOGE)	Currency
5	Dogelon Mars (ELON)	Currency
6	Litecoin (LTC)	Currency
7	Magic Internet Money (MIM)	Currency
8	Pax Gold (PAXG)	Currency
9	Shiba Inu (SHIB)	Currency
10	Circle Coin (USDC)	Currency
11	Zcash (ZEC)	Currency

Table 3: Category: Currencies

	Crypto Asset	Category
12	Ethereum (ETH)	L1/L2
13	Fantom (FTM) (+)	L1/L2
14	Keep3rV1 (KP3R)	L1/L2
15	Terra (LUNA)	L1/L2
16	Polygon (MATIC)	L1/L2
17	Quant (QNT)	L1/L2
18	Republic Protocol (REN)	L1/L2
19	Skale Network (SKL)	L1/L2
20	Solana (SOL)	L1/L2
21	Tezos (XTZ)	L1/L2

Table 4: Category: L1/L2 Platforms

	Crypto Asset	Category
22	1inch (1INCH)	DeFi
23	AAVE (AAVE)	DeFi
24	Alchemix (ALCX)	DeFi
25	Balancer (BAL)	DeFi
26	Bancor (BNT)	DeFi
27	BarnBridge (BOND)	DeFi
28	Compound (COMP)	DeFi
29	Curve(DAO) (CRV)	DeFi
30	Cryptex Finance (CTX)	DeFi
31	Injective Protocol (INJ)	DeFi
32	Kyber Network (KNC)	DeFi
33	Loopring (LRC)	DeFi
34	Moss Carbon Credit (MCO2)	DeFi

Table 5: Decentralized Finance

	Crypto Asset	Category
35	Flexa Network (AMP)	WEB3
36	ANKR (ANKR)	WEB3
37	API3 DAO (API3)	WEB3
38	Audius (AUDIO)	WEB3
39	Axie Infinity (AXS)	WEB3
40	Basic Attent Token (BAT)	WEB3
41	Civic (CVC)	WEB3
42	Enjin Coin (ENJ)	WEB3
43	Ethereum Name Service (ENS)	WEB3
44	Fetch.Ai (FET)	WEB3
45	Filecoin (FIL)	WEB3
46	Gala Games (GALA)	WEB3
47	Teh Graph (GRT)	WEB3
48	Chainlink (LINK)	WEB3
49	Livepeer (LPT)	WEB3
50	Mask Network (MASK)	WEB3
51	Merit Circle DAO (MC)	WEB3
52	Orchid (OXT)	WEB3
53	Radicale (RAD)	WEB3
54	Render Token (RNDR)	WEB3
55	Smooth Love Potion (SLP)	WEB3
56	Storj (STORJ)	WEB3

Table 6: Web 3.0

	Crypto Asset	Category
57	Somnium Space (CUBE)	Agg/Meta
58	Decentraland (MANA)	Agg/Meta
59	The Sandbox (SAND)	Agg/Meta

Table 7: Aggregators/Metaverse